IVAN JORGE CABALLOS ESPINOSA

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PROFILE

Graduate Electrical & Electronic Engineer with hands-on experience in production programming, functional test execution, and first-line PCB fault finding in an EMS environment. Strong focus on hardware verification and bring-up: integration, measurement-driven debugging, root-cause isolation, and clear documentation. Comfortable working to controlled procedures, capturing test data for traceability, and using bench instruments (PSU, DMM) to verify electrical behaviour and isolate faults. Portfolio includes measured hardware work (constant-current driver) and embedded systems (ESP32-S3). Targeting a Graduate role in hardware integration / verification / test-debug, with a longer-term interest in analog and embedded design.

KEY TECHNICAL SKILLS

- Hardware verification & bring-up: board integration, structured debug, rail checks, RCA/5-Why thinking, clear test notes
- Production test execution: pass/fail criteria, controlled procedures, repeatable station operation, traceability mindset
- PCB troubleshooting: visual inspection (placement/solder/BOM), continuity/short checks, missing-rail isolation using DMM
- Embedded systems (project work): ESP32-S3, Embedded C/C++ (Arduino framework), I2C/SPI, Wi-Fi (NTP), ESP-NOW
- Software/tools: Python (Flask, systemd services), Git/GitHub, GitHub Pages documentation, basic Linux/Raspberry Pi
- Programming & production tools: Flash Development Toolkit (Renesas), OPTWIN, MPLAB (Microchip)
- Lab instrumentation: bench PSU, bench/handheld DMMs; basic oscilloscope exposure (as required on bench)

PROFESSIONAL EXPERIENCE

Test Technician — Dynamic EMS | Dalgety Bay, Scotland | Nov 2025 – Present

- Program PCB assemblies via station PCs, flash production images and validate correct configuration using FDT / OPTWIN / MPLAB before test.
- Execute functional test procedures: power boards using a bench PSU at specified conditions, verify outcomes against pass/fail criteria, and capture results for traceability.
- Maintain throughput of ~25–30 boards/day while keeping test execution consistent and records accurate.
- Troubleshoot failed units: visual checks (incorrect/missing components, solder quality) and DMM checks to identify missing rails, shorts, and continuity issues.

- Operational continuity: when a legacy Windows XP programming station failed, reconfigured an alternative station setup (PC + auxiliary power board) to minimise downtime and keep production moving.

Royal Mail — Postman (previous / part-time)

- Time-critical delivery, reliability, and consistent execution to standard procedures under daily throughput.

EDUCATION

BSc Electrical & Electronic Engineering — Edinburgh Napier University (Awarded Oct 2023)

HNC Electrical Engineering — Glasgow Kelvin College

SELECTED PROJECTS (PORTFOLIO)

Links: ivan-caballos.github.io | linkedin.com/in/ivan-caballos-461193147

Laser Constant-Current Driver Prototype (0–200 mA target) — measured validation

- Bench-tested constant-current driver; compared theory vs simulation vs measurement and iterated wiring/power integrity to improve stability.
- Captured measurement tables and documented conclusions, including instrumentation effects (ammeter burden voltage) and next design steps.

Clock and Weather Station — ESP32-S3 (ILI9341 + AHT20 + BMP280, NTP + Open-Meteo, battery-powered)

- Integrated TFT UI + indoor sensing + online forecast; implemented UK NTP time sync (GMT/BST) and on-demand Wi-Fi updates.
- Added battery monitoring (ADC + calibration/smoothing) and documented bench PSU current (~40 mA boot, ~15 mA idle with display on).
- Documented schematic and PCB iterations with a serviceable layout direction for final assembly/enclosure.

Manufacturing Process Improvement (Case Study) — Test/Programming Automation Roadmap (sanitised)

- Created a phased roadmap to improve repeatability and traceability (QR logging, variant profiles, incremental automation).
- Focused on baseline measurement first to enable later ROI/cycle-time quantification from real data.

Other projects: AI Lab Assistant (Raspberry Pi, Flask, LAN-only engineering log); 555 monostable timer PCB (measured); binary/decimal counters (logic fundamentals).